

MAC 233: External LED Lab

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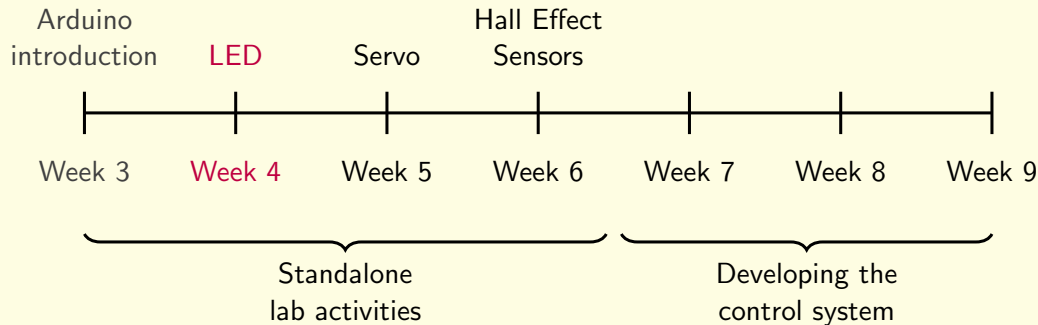
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MAC 233: **21/10/2026**



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MAC 233: LABS OVERVIEW



TODAY'S LAB: PREREQUISITES

This worksheet builds upon the Lab 1 worksheet. You will have needed to complete the Lab 1 worksheet before you can continue with this lab.

Out of your Arduino pack, you will need: 1 Arduino, 1 breadboard, 1 LED, 1 220Ω resistor, and 1 jumper cable.

TODAY'S LAB: OBJECTIVES

The objective of today's lab is to learn how to use an electronic breadboard to create a circuit and to control an external output on the Arduino.

We will be taking the scenario from the previous worksheet and changing it to communicate with an external LED.

REMINDER: 'PINMODE' AND 'DIGITALWRITE'

pinMode sets the IO direction of a pin and belongs in setup.

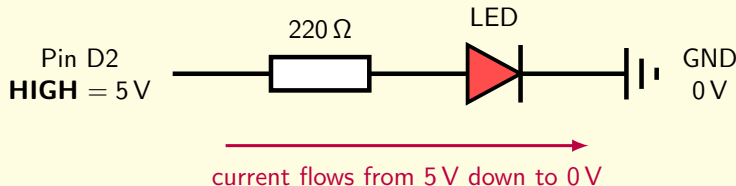
digitalWrite then sets the pin HIGH (+5V) or LOW (0V).

```
#define LED_PIN 2
```

```
void setup() {  
    pinMode(LED_PIN, OUTPUT);    // once, in setup  
}
```

```
void loop() {  
    digitalWrite(LED_PIN, HIGH); // 5 V, LED on  
    delay(1000);  
    digitalWrite(LED_PIN, LOW);  // 0 V, LED off  
    delay(1000);  
}
```

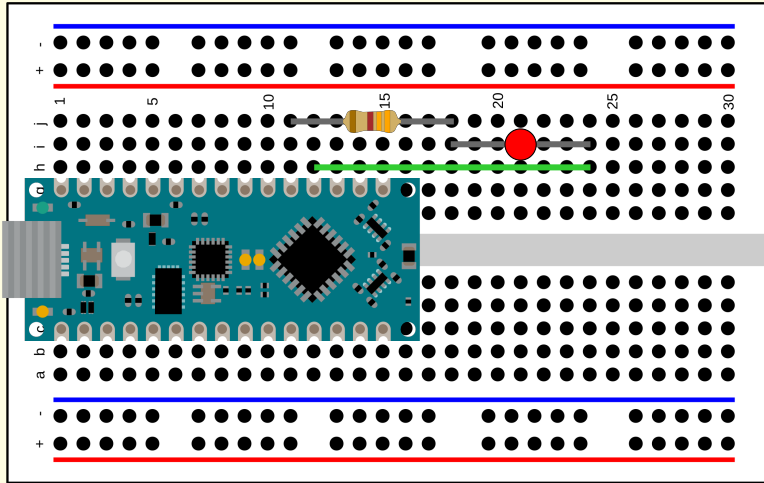
WHAT 'HIGH' ACTUALLY DOES



`digitalWrite(LED_PIN, HIGH)` connects the pin to 5V, and 'LOW' connects it to 0V. It is that 5V that pushes a current through the LED and lights it.

The LED itself uses up about 2V, so the resistor takes the other 3V. With no resistor, nothing limits the current and both the LED and the pin can be damaged.

WIRING DIAGRAM



Breadboard source: <https://freesvg.org/breadboard>, Resistor source <https://freesvg.org/resistor-330-ohm>.